

The physics and mathematics of ziplining

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Abstract

We study the kinematic and dynamic characteristics of body motion along an inclined hanging cable in the approximation of its masslessness and inextensibility. It is shown that in this case, the motion of the suspended body occurs along an inclined ellipse, the foci of which are the cable attachment points. Herewith, at any given moment, the net tension force of the cable is perpendicular to the instantaneous velocity of the body. Besides, the tension force is a dynamic characteristic of the cable and increases as the body descends along with the increase in the speed. There is also a sharp increase in the tension at the end of the trajectory, which is explained by a decrease in its radius of curvature.

Keywords: zip-line, massless cable, ellipse, tension force, holonomic constraint.

1 Introduction

Zip-line (also called the "tyrolean" or "hanging cable") is a pulley suspended on an inclined cable made of stainless steel [1, 2, 3]. This design is intended to allow a load or person to move (due to the gravity force) from the top to the bottom of the cable, holding on or being attached to a free-moving pulley

(figure 1). Its application is not limited to adventure sports, recreation, or tourism, although modern usage tends to favor these species.



Figure 1: The motion along the zip-line cable car.

Similar devices, consisting of cables or ropes and blocks located between two points, which use the force of gravity for transportation, have been known to people since ancient times. They were often used in rugged terrain to transport people and cargo through canyons, rivers, and impassable places in remote areas of China, the Costa Rican rainforest, and the Australian outback.

Such mechanical structures are called differently in various countries: in the Alps, they are called “Tyrolean crossing” or “Tyrolean traverse”, in other places they are known as “Flying fox” or “inclined strong”. In H.G. Wells’s 1897 novel “The Invisible Man”, the “inclined strong” is mentioned as part of “All Saints’ Day”.

Mungan and Lipscombe [4] studied the trajectory of a rider slowly moving along a zip-line of a fixed length. They showed that by increasing the relative mass of the rider (relative to the total mass of the cable) from zero to infinity, the motion trajectories could be uniformly interpolated between a catenary line and an ellipse.

The paper [5] describes the procedure for creating a mathematical model of descending a rider down a heavy rope that takes into account mechanical and aero-dynamical resistances that occur during the descent. The analytical investigation of the oscillations of an elastic cable loaded with a mass point is presented in Ref. [6]. The Ref. [7] defined the determination of zip-line

passengers braking distance. The paper [8] discussed a dynamic engineering problem of a mass attached to a pendulum sliding along a cable.

In this work, we study the kinematic and dynamic characteristics of such motion in the approximation of a massless inextensible cable. The issue is best suited for the beginning of an upper-level undergraduate course in classical mechanics. This topic may be recommended for further reading, and can be used in physics electives or as an undergraduate project.

2 The basic model assumptions

Let us consider a homogeneous cable fixed at two stationary points, the shortest distance between which is equal to f . We take that these two points are separated vertically by a certain distance (height difference) so that the angle of inclination of the line segment connecting these two points to the horizon (slope angle) is acute and equal to α (figure 2).

We also assume that the total cable length remains constant and equal to l . Additionally, we suppose that in the free state, the cable is not tensioned (slacks) so $l > f$. We need to determine the speed of a body v that can freely slide down along it under the influence of gravity and the tension force of the cable $\vec{T}$ that arises in this case if the body mass is equal to m .

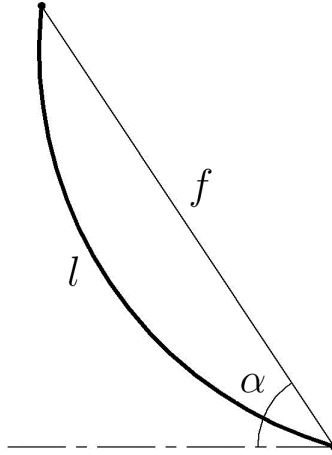


Figure 2: Geometry of the problem.

We consider the cable to be a massless, flexible, inextensible, and homogeneous thread. The condition that the cable is massless means that its mass

is much less than the mass of the suspended body. In other words, at any moment of movement of the body, the total mechanical energy of the cable can be neglected in comparison with the mechanical energy of the body. This approximation is correct in many practical cases, especially for small distances f . As an example, we consider the recommended 3/16" galvanized aircraft cable [1] with linear density equal to 0.06 kg/ft. Then, the mass of 100 ft zip-line makes only 6 kg which is much less than the average mass of an adult person.

The flexibility indicates that the cable has infinitesimal bending rigidity and is capable of "working" only in tension. This condition may be achieved only when the load has a significantly larger mass compared to the zip-line. The inextensibility condition assumes that the total length of the cable is constant for an arbitrary weight of the load. Thus, when the body moves, the total deformation energy of the cable can be neglected compared to the mechanical energy of the body. Additionally, we neglect the air resistance to the body motion and the rolling friction of the pulley.

If a body is suspended from a massless cable at some point, then such a cable will stretch, forming two line segments with a cusp at the pulley (the other two ends of these segments correspond to the cable fixation points). However, since the cable is inextensible, the distance from the fixed points to the body at any point in the trajectory of its movement remains constant and equal to l . But this condition is the main property of an ellipse with foci F_1 and F_2 located at these two points and the distance between focal points F_1 and F_2 is f . Therefore, in this approximation, the body will move along the section of the ellipse (figure 3)!

3 Finding the instantaneous speed

Firstly, we note that the limited motion of the body discussed above is called constrained motion. However, due to the neglect of the mechanical energy of the cable and its deformation energy, the total mechanical energy of the suspended body must be conserved in time.

Now, we introduce a two-dimensional coordinate system as shown in figure 3. The vertical height h shown in figure 4 is measured from the origin O (the center of the ellipse). Using the law of energy conservation, we get:

$$v = \sqrt{2g(h_0 - h)}, \quad (1)$$

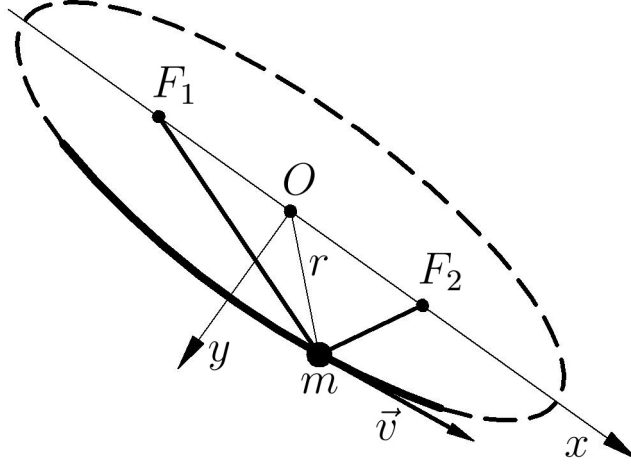


Figure 3: The idealized trajectory of the body motion (the ellipse section).

where h_0 is the initial height; g is the acceleration of gravity. It follows from figure 4 that

$$h = \pm r \sin(\alpha + \varphi), \quad (2)$$

where the sign " + (-) " corresponds to negative (positive) values of the x coordinate; $r = \sqrt{x^2 + y^2}$ (the radial coordinate); $\varphi = \arctan(y/x)$ (the polar angle).

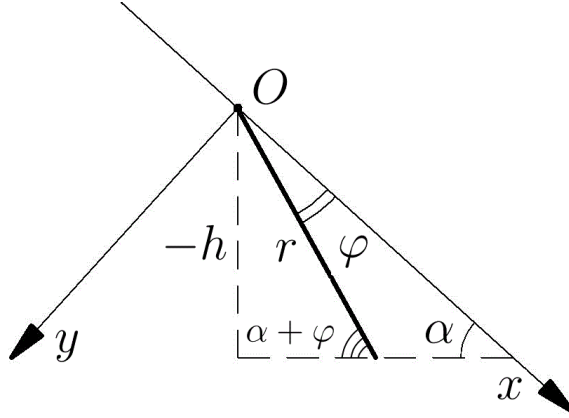


Figure 4: The diagram explaining equation (2).

As is known, the equation of an ellipse has the form:

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, \quad (3)$$

where $a = l/2$ is the length of the semi-major axis of an ellipse; $b = \sqrt{l^2 - f^2}/2$ is the length of the semi-minor axis of an ellipse. Solving this equation with respect to y coordinate, we derive:

$$y = \sqrt{\frac{l^2 - f^2}{4} - \frac{l^2 - f^2}{l^2} x^2}. \quad (4)$$

In order to reduce the number of input parameters of the problem (from three to two) and to compact the final equations, we introduce the following set of dimensionless (reduced) variables: $\tilde{v} = v/\sqrt{gl}$ (the dimensionless speed), $\tilde{x} = x/l$ (the dimensionless longitudinal coordinate). The maximum allowable range of $\tilde{x}$ values required to hold the load on the ellipse is $-0.5 \leq \tilde{x} \leq 0.5$. Indeed, the points $x = \pm a = \pm l/2$ correspond to two opposite vertices of the ellipse. This means that $|\tilde{x}| \leq 0.5$.

We also introduce the dimensionless parameter $k = f/l \leq 1$, which characterizes the degree of slacking of the cable. Then, taking into account equations (1), (2), (4), we obtain:

$$\tilde{v} = \sqrt{\psi_0 - \psi}, \quad (5)$$

where

$$\psi(\tilde{x}) = \pm \sqrt{1 + k^2(4\tilde{x}^2 - 1)} \sin \left[\alpha + \arctan \left(\frac{\sqrt{(1 - k^2)(1 - 4\tilde{x}^2)}}{2\tilde{x}} \right) \right] \quad (6)$$

and $\psi_0 = \psi(\tilde{x}_0)$. Therefore, the body speed $\tilde{v}$ is a function of the coordinate $\tilde{x}$ and depends on two parameters: α and k .

Figure 5 shows the dependence $\tilde{v}(\tilde{x})$ constructed for different values of parameters α and k . It is seen that with an increase in the angle of inclination and a decrease in the cable slacking coefficient, the values of the instantaneous speed increase slightly.

4 Finding the tension force of the cable

At any given moment, the suspended body is acted upon by the tension forces of the cable $\vec{T}_1$ and $\vec{T}_2$ and the vertically downward gravity force $m\vec{g}$

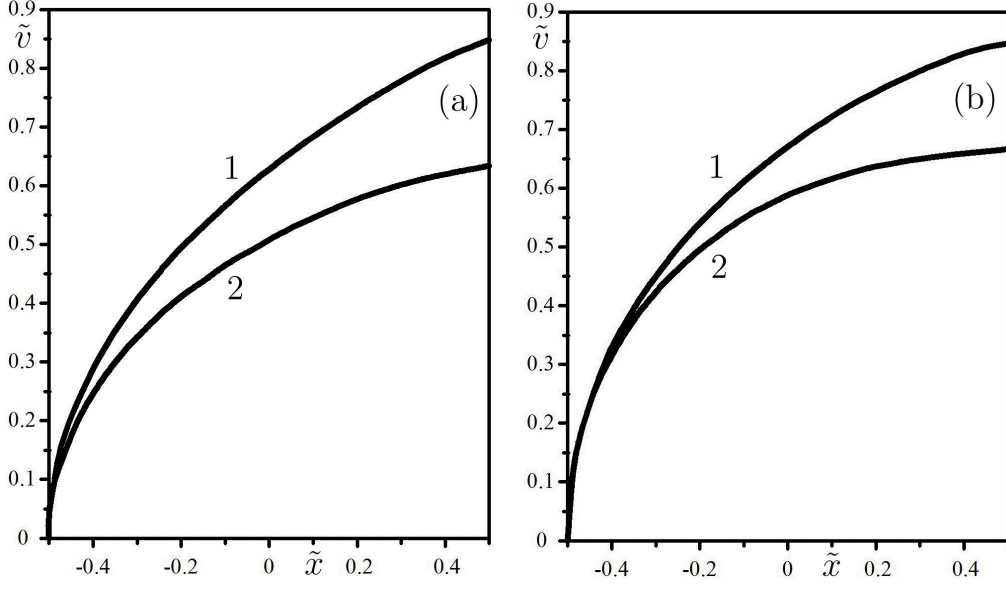


Figure 5: Dependence $\tilde{v}(\tilde{x})$ at $\alpha = 60^\circ$ (1) and $\alpha = 30^\circ$ (2). (a): $k = 0.94$; (b): $k = 0.54$.

(figure 6). Assuming the rotating part of the pulley is both massless and have a frictionless axle of rotation, these two tension forces should be the same in magnitude: $T_1 = T_2 = T$. Given the reflective property of an ellipse (the normal $\vec{n}$ to any point of the ellipse forms equal acute angles with the focal radii of this point), we have: $\beta_1 = \beta_2 = \beta$.

Thus, the net tension force is perpendicular to the instantaneous velocity of the body! Therefore, it does not perform work and the change in the absolute value of the body's velocity occurs only due to the gravity, which was already used when finding its speed in the previous section. In the scientific literature, such a mechanical constraint is called an ideal holonomic constraint. Interestingly, given the model assumptions, the original problem is completely equivalent to the problem of a body sliding along the smooth inner surface of an ellipsoidal cylinder in a gravitational field. In that case, the net tension force represents the normal force of the support of such a surface.

In the light of the foregoing consideration, Newton's second law in the

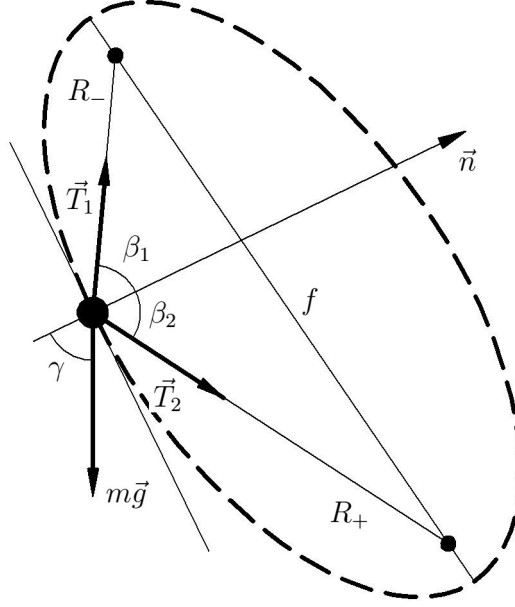


Figure 6: Forces acting on the body during its motion along a zip-line. Here, $\vec{n}$ is the vector perpendicular to the tangent line and directed towards the concavity of the ellipse.

direction of the normal to the trajectory takes the form:

$$\frac{mv^2}{\rho} = 2T \cos \beta - mg \cos \gamma, \quad (7)$$

where ρ is the absolute value of the instantaneous radius of curvature of the trajectory; γ is an acute angle between the gravity and the normal. Using equation (7), we find:

$$\tilde{T} = \frac{\cos \gamma + \tilde{v}^2 / \tilde{\rho}}{2 \cos \beta}, \quad (8)$$

where $\tilde{T} = T/mg$ is the dimensionless tension force of the cable; $\tilde{\rho} = \rho/l$ is the dimensionless radius of curvature.

The absolute value of the radius of curvature of a plane curve, which is given by the equation $y(x)$, can be found by the well-known equation [4]:

$$\rho(x) = \frac{(1 + y'^2(x))^{3/2}}{|y''(x)|}, \quad (9)$$

where $y'(x)$ and $y''(x)$ are, respectively, the first and second derivatives of function $y(x)$. Using equations (4) and (9), we get:

$$\tilde{\rho}(\tilde{x}) = \frac{(1 - 4k^2\tilde{x}^2)^{3/2}}{2\sqrt{1 - k^2}}. \quad (10)$$

In order to find angle β as a function of coordinate x , we use the cosine theorem (see figure 6):

$$\cos 2\beta = \frac{R_-^2 + R_+^2 - f^2}{2R_-R_+}, \quad (11)$$

where R_- and R_+ are the smaller and larger focal radii of the ellipse point with coordinates (x, y) (figure 6). On the other hand, applying the Pythagorean theorem, we derive:

$$R_{\pm}^2 = \left(\frac{f}{2} \pm x\right)^2 + y^2. \quad (12)$$

Taking into account equations (4), (11), (12), and the well-known relation $\cos \beta = \sqrt{(1 + \cos 2\beta)/2}$, we obtain:

$$\cos \beta(\tilde{x}) = \sqrt{\frac{1 - k^2}{1 - 4k^2\tilde{x}^2}}. \quad (13)$$

It follows from equations (8) and (13) that at $k \rightarrow 1$ (the cable is not slacked) the tension force increases indefinitely. Consequently, in this limiting case, our model assumption about the inextensibility of the rope is not correct, and we need to take into account additionally its deformation arising under the influence of its own weight and the weight of the suspended body.

Using figure 7, we obtain:

$$\gamma = \alpha + \theta. \quad (14)$$

where θ is the angle between the instantaneous velocity and x -axis. But

$$\tan \theta = y'(x). \quad (15)$$

Then, using equations (4), (14), and (15), we get:

$$\cos \gamma(\tilde{x}) = \cos \left[\alpha - \arctan \left(2\tilde{x} \sqrt{\frac{1 - k^2}{1 - 4\tilde{x}^2}} \right) \right]. \quad (16)$$

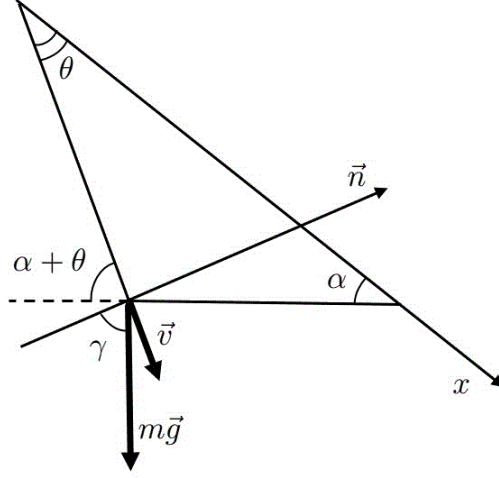


Figure 7: The diagram explaining equation (14).

Figure 8 shows dependence $\tilde{T}(\tilde{x})$ constructed for different values of parameters α and k with the help of equations (5), (6), (8), (10), (13), (16). It is seen that the tension force is a dynamic characteristic of the cable and increases as $\tilde{x}$ increases, which is due to the increase in the speed of the body during its motion. A sharp increase in $\tilde{T}$ at the end of the trajectory is explained by a decrease in its radius of curvature. As α increases, the cable tension at the end of the motion also increases.

5 Discussion

The main result of the present work (which distinguishes it from the conclusions available in previous literature) should be considered the identification of a sharp increase in cable tension at the end of the load's traveling. This hypothesis is very important due to the problem of insuring the motion of the cable car users because of a possible cable break.

The reader can intuitively assume in advance that the tension force of the zip-line should increase as the body moves. The constructed dynamic model allows us to confirm this assumption. Moreover, it predicts that at the initial stage of motion, the tension has a larger magnitude for smaller inclination angles (curves 1 in Figure 8). On the contrary, at the end of the motion, the tension is greater for large α (curves 2 in Figure 8). These two circumstances

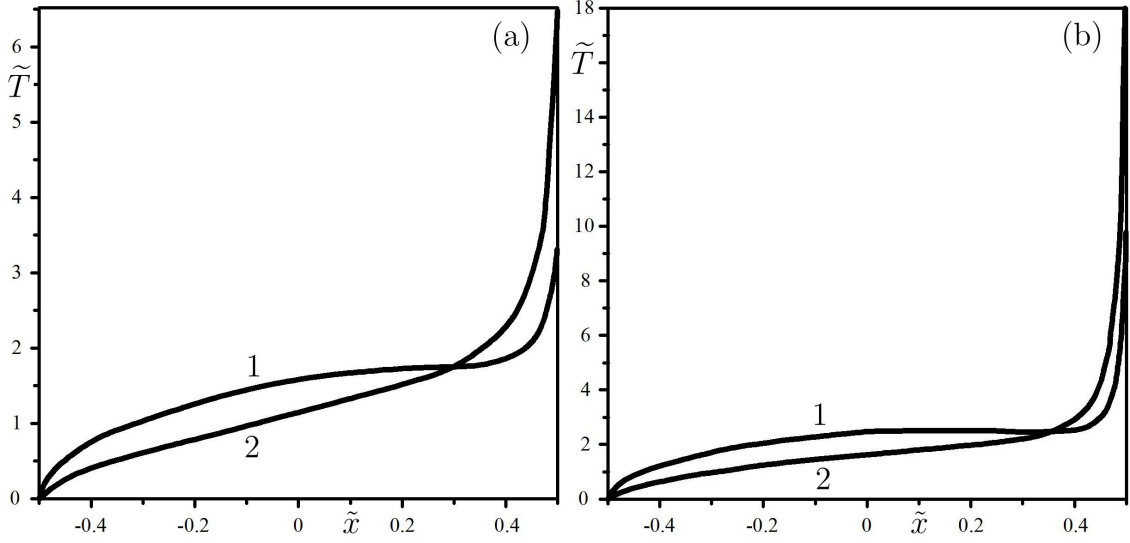


Figure 8: Dependence $\tilde{T}(\tilde{x})$ at $k = 0.94$ (a) and $k = 0.98$ (b). (1): $\alpha = 30^\circ$; (2): $\alpha = 60^\circ$.

inevitably lead to the fact that curves 1 and 2 in Figure 8 should intersect at a certain value of $\tilde{x}$.

To our knowledge, there is no available experimental data concerning the change of the tension force during the body motion (the presented in Ref. [5] measurement results relate only to the time dependence of the rider's speed). In this regard, we may propose an experimental verification of the numerical results obtained in this work as an interesting undergraduate project or even as a research paper.

6 Conclusions

The above analysis has been carried out in the simplest possible approximation of a massless inextensible cable (including neglect of the aerodynamic drag, as well as the rolling friction in the pulley) and can be proposed to undergraduate students. Our calculations involve important concepts and techniques, including the elements of kinematics and dynamics of curvilinear motion, analytic geometry, differential calculus, and computational methods. Applying these techniques to a practical problem illustrates their applicabil-

ity and engages the interest of students perhaps more than would trivial school problems.

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